



***NovaSource G6™* RF Signal Source**

User Manual

Revision 1.2

**L-3 Communications
Nova Engineering**

Table of Contents

Table of Contents.....	ii
Introduction	4
Installing <i>NovaSource G6</i>	6
Package Contents.....	6
Set Up.....	7
Install Hardware	7
Install NovaSource G6 Control Panel Software (NSCP G6)	8
Operation Via PC Interface.....	10
Control Panel Window.....	11
Menus	12
File Menu	12
Settings	13
COM Port.....	15
Help.....	16
Selecting and Changing Controls.....	16
Frequency Controls.....	17
Attenuation	20
Trigger Control	22
Selecting the Trigger Source.....	22
Continuous, Momentary, Toggle.....	23
RF Standby Mode	24
Function Buttons	26

Read	26
Load and Store	27
Trigger	27
LED Display.....	28
Status Display	30
Modulation Controls.....	30
Modulation Source Selection	31
Modulation On/Off Control	31
Deviation Control	32
Sweep Window	33
How NovaSource G6 Works	36
Power	38
About Nova Engineering	39
Getting Help.....	39
Copyright	40
Trademarks	41
Nova Engineering Product Warranty	42
NovaSource G6 Specifications.....	44
Index.....	47

Introduction

Nova Engineering's *NovaSource G6* Programmable Synthesized RF Signal Sources are completely self-contained, with internal TCXO reference, and software controlled attenuator.

With your PC and the supplied *NovaSource G6 Control Panel* software, you simply enter the desired frequency and select an RF output mode and power level. Your settings can be stored in non-volatile memory, so once configured, *NovaSource G6* is "plug and play."

Additionally, the internal reference signal can be brought out of the *NovaSource G6* or an external reference signal can be used.

The modulation feature on the *NovaSource G6* allows either an externally or internally generated signal to modulate the RF output signal. Therefore, *NovaSource G6* can replace full size signal generators in a wide variety of applications.

NovaSource G6 can also be continuously controlled via its serial input, perfect for product development, breadboarding, test applications, general lab use, or small run production.

Models are available in all popular bands, from 120 MHz to 5.875 GHz, with 1 kHz tuning resolution.

Nova Engineering's *NovaSource G6 RF Signal Source* is designed specifically to fill the void for a small, low cost, and easy to use signal generator. *NovaSource G6* is an alternative to more expensive and cumbersome signal generators which have features not necessary for many projects. For applications in which speed, flexibility, and accuracy are required, *NovaSource G6* is ideal.



Figure 1: NovaSource G6 RF Signal Source

NovaSource G6 is a complete solution for many frequency synthesizer applications. Requiring only DC power, *NovaSource G6* can be used right out of the box with the unit's built-in settings. Using the supplied software, *NovaSource G6* settings can be changed and controlled from any Windows-based PC.

Each standard *NovaSource G6* includes all synthesizer functions, including a buffered RF output and an internal ± 1 PPM Temperature Compensated Crystal Oscillator (TCXO) reference. The reference port connector can be used to output the internal TCXO signal or to input an external reference signal, making *NovaSource G6* ideal for use in a system application. Internal or external mode is selectable via a front panel mounted switch.

Finally, either an external on/off trigger or an analog modulation signal can be supplied to *NovaSource G6*. Three trigger modes are available for controlling the RF output state. An RF standby mode provides quick RF output on/off switching. An analog waveform can be applied, or internal modulation selected, to create various types of FM modulation.

Installing *NovaSource G6*

NovaSource G6 requires no special equipment and is ready to use right out of the box. *NovaSource G6* can be used alone, or host controlled with the accompanying *NovaSource G6 Control Panel* (NSCP G6) software or any terminal program. A spectrum analyzer or frequency counter may be used to monitor the RF output, if desired.

Package Contents

Your *NovaSource G6* package includes the following:

Table 1: *NovaSource G6* Package Contents

Item	Description	Part Number
<i>NovaSource G6</i>	RF Signal Source	NS3-ffffsss (where ffff is the lowest frequency in MHz and sss is the smallest step size in kHz)
<i>NovaSource G6 Control Panel Software</i>	Supplied on CD-ROM	NL-CDR01-C
PC Interface Cable	9 pin male x 9 pin female	8051-0115
Wall Plug Power Supply	9 VDC @ 500mA UL/CSA (North American models only)	NS3-WPPS
Spare Power Supply Connector	3.5mm x 1.3mm Female Barrel	W00

Set Up

NovaSource G6 may be set up as a dedicated stand-alone unit (program it at startup, disconnect from the PC, and run indefinitely) or as a programmable PC peripheral. The *NovaSource G6* settings are adjusted and controlled using the included *NovaSource G6 Control Panel* (NSCP G6) software. This software interface can be part of an application that continually changes the *NovaSource G6* settings. The NSCP G6 software can also be used to configure *NovaSource G6* for applications that require less frequent changes. For detailed information about the interface settings and command set, refer to Nova Application Note "Programming the *NovaSource G6*" available from our website at <http://www.nova-eng.com>.

Install Hardware

This section provides a step-by-step procedure that explains how to set up and operate *NovaSource G6* as a stand-alone unit using the *NovaSource G6* default settings.

- Plug the included DC power supply cable into the power supply receptacle located on the *NovaSource G6* rear panel.
- Plug the wall mount power supply unit into a 110 VAC outlet. (North American models)

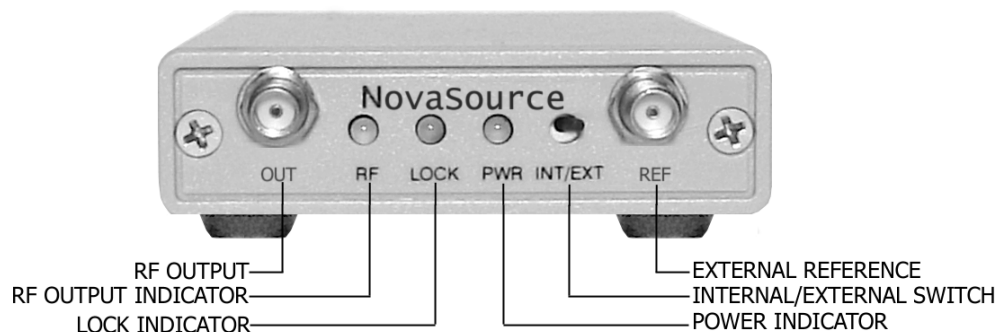


Figure 2: NovaSource G6 Front Panel

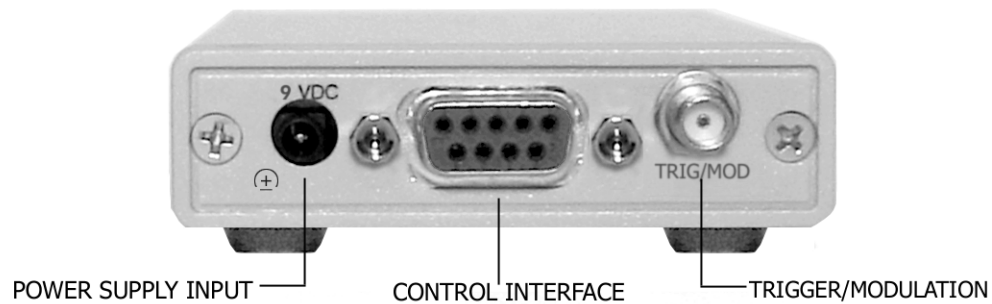


Figure 3: NovaSource G6 Rear Panel

The power LED illuminates immediately. The LOCK and RF LEDs should also display. This indicates that *NovaSource G6* is locked on frequency and is producing a signal. *NovaSource G6* is now ready to be used as a stand-alone unit.

Note: The default power-on frequency for *NovaSource G6* is its minimum frequency.

The INT/EXT switch on the front panel of the *NovaSource G6* can be used to set the unit to operate from an external reference. When the switch is in the INT position, the internal 10 MHz TCXO is used as the reference. The internal reference signal will be present on the REF connector. When the switch is in the EXT position, the REF connector becomes an input. An external reference signal can then be applied. Note that the reference frequency, internal or external, must match the frequency provided in the RF command. The default is 10 MHz.

Install NovaSource G6 Control Panel Software (NSCP G6)

The *NovaSource G6 Control Panel* (NSCP G6) software included with your *NovaSource G6* package is a Windows 95/98/NT/2000 application which may be used to configure and control *NovaSource G6*. The following system configuration is required as a minimum in order to use NSCP G6:

- PC running Windows 95/98/NT/2000/XP
- 16MB RAM

- 5MB available hard disk space
- CD-ROM drive
- 640 x 480 resolution color or grayscale video monitor
- 9-pin female D-Sub, Serial (COM) port

This section provides a step-by-step procedure that explains how to set up and operate *NovaSource G6* using the NSCP *G6* software.

- Using the included 9 pin serial cable, connect the *NovaSource G6* to the serial port of a PC running Windows.
- Insert the included CD into the CD-ROM drive of the host computer.
- From the CD, select the folder labeled NS_G6.
- Run SETUP.EXE, and follow the on-screen instructions.



Figure 4: NovaSource G6 Rear Panel

- The NSCP *G6* installer decompresses and copies all of the files to their proper locations. The installer also creates a program group containing the NSCP *G6* icon.

Your *NovaSource G6* is now ready to be used with the NSCP *G6* software.

Operation Via PC Interface

The NSCP G6 software program consists of two main windows: Control Panel and the Sweep Window.

The Control Panel Window is used to:

- Set the frequency
- Set the external input mode
- Set the RF output trigger mode
- Set the Modulation controls
- Set the RF attenuation
- Load and store configurations

The Sweep Window is used to set up and control sweeps between two end point frequencies.

NSCP G6 settings are retained between sessions in the application's initialization file.

Control Panel Window

The Control Panel Window is the main NSCP G6 window and is displayed when the program is launched. This window contains: Menus, Controls, Displays, Modes, Buttons, and the Status Display, each of which is described below.

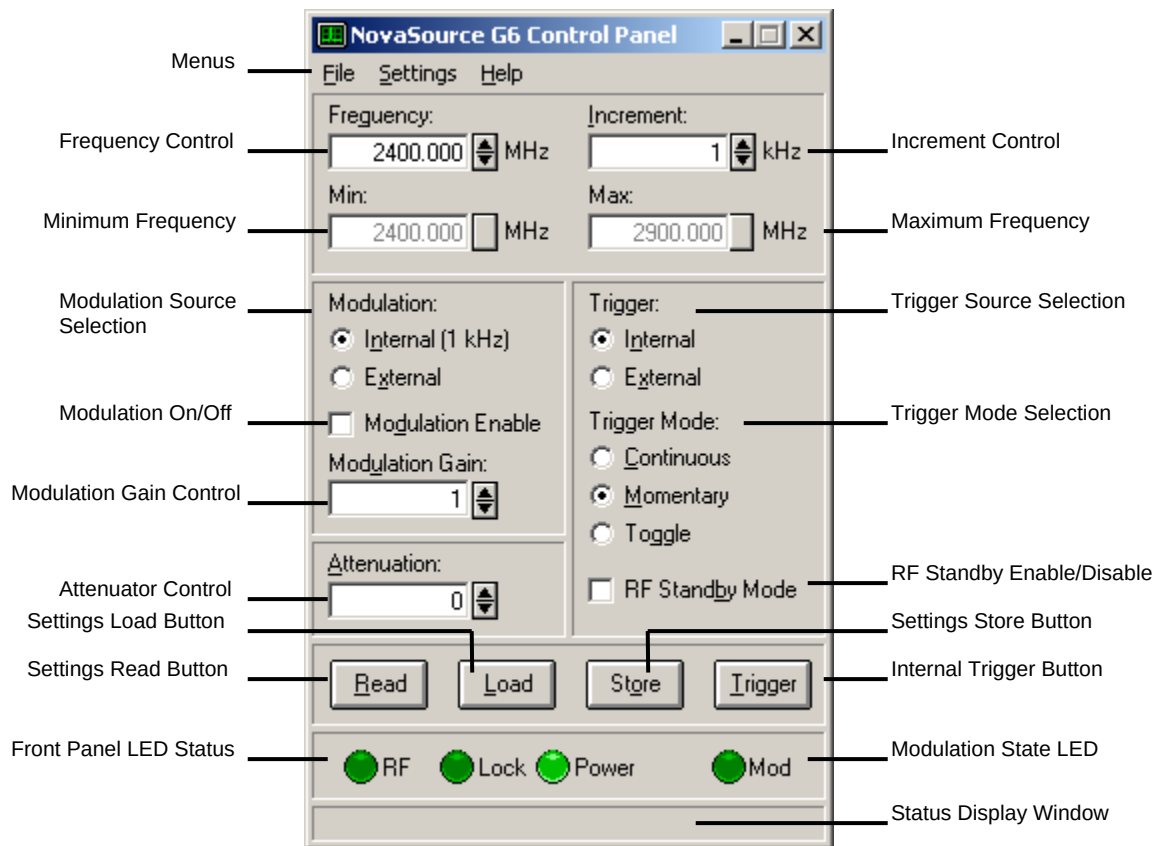


Figure 5: NovaSource G6 Control Panel Main Window

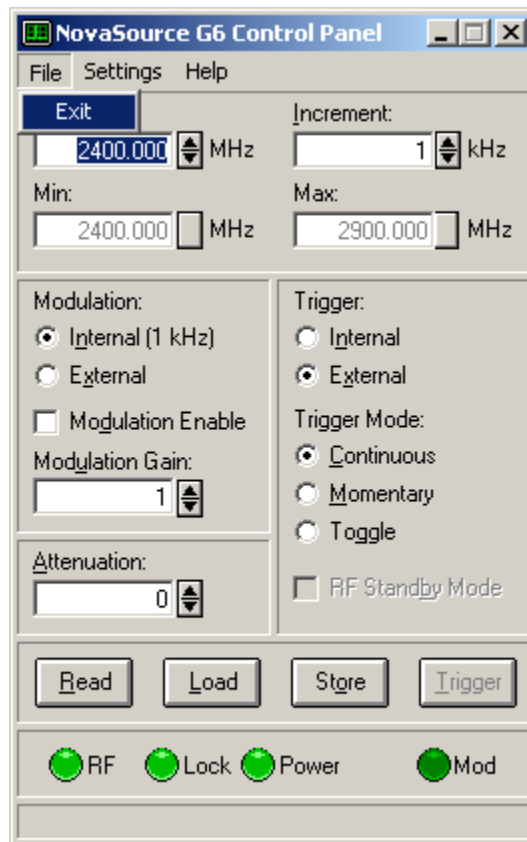
Menus

There are three menus in the NSCP G6: File, Settings, and Help. These menus are used to configure the NSCP G6, to launch the Sweep Window, and for getting help.

File Menu

The File Menu provides the means for exiting the application. To close the NSCP G6, select File, then select Exit. All of the current settings for the Control Panel Window and the Sweep Window will be saved in the initialization file on the computer.

Control Panel Window, File Menu

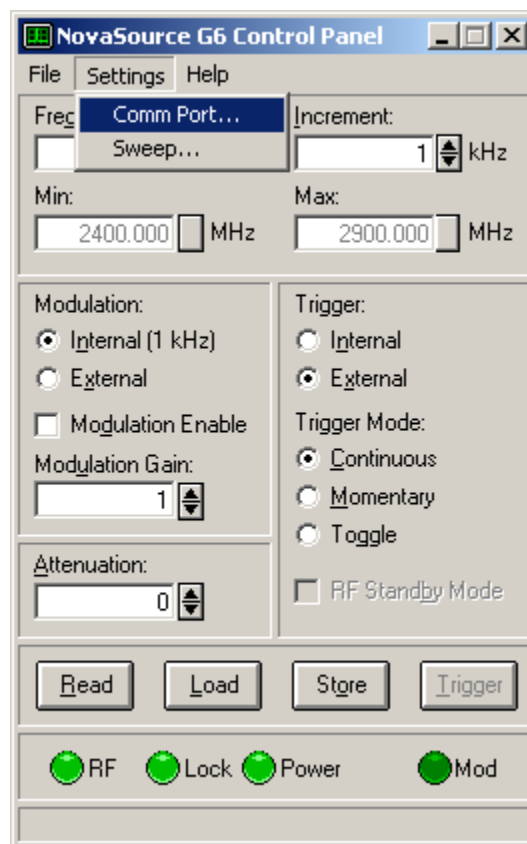


Field Name	Description
Exit	Prompts to store the current settings and terminates the NSCP G6 application.

Settings

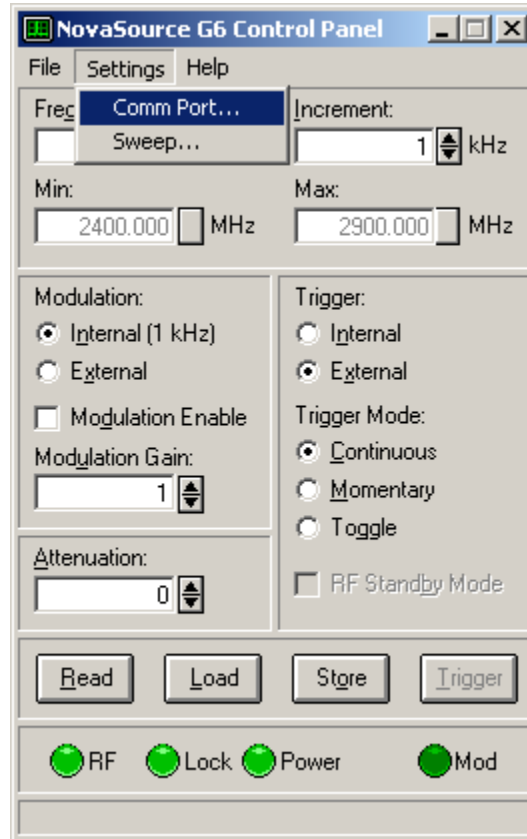
The Settings Menu is used to change system settings.

Control Panel Window, Settings Menu



Field Name	Description
Comm Port	Launches a dialog box used to select the serial port to which <i>NovaSource G6</i> is connected.

Control Panel Window, Settings Menu



Field Name	Description
Sweep	Opens the Sweep Window. Sweep sends commands to NovaSource G6 to sweep across frequencies between two user-selected limits at the user-selected rate and step-size.

COM Port

When you click on the down arrow in the Serial Port dialog box, a dropdown menu displays a list of communication ports.

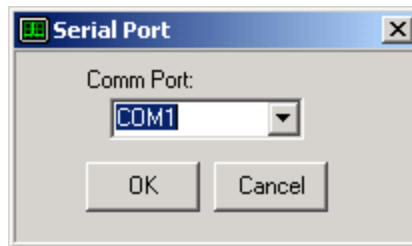


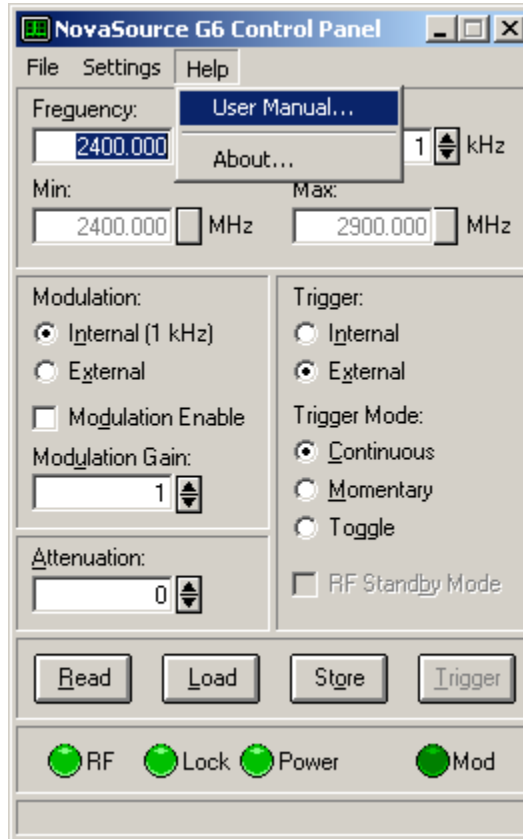
Figure 6: Serial Port Dialog Box

Click on the OK button after choosing a port address (COM1, COM2, COM3, etc.). This sets the new port address for the NSCP G6 to communicate with *NovaSource G6*.

Clicking on Cancel or closing the Serial Port window restores the previous address. All other settings for the COM port, such as baud rate, parity, etc. are preset by the NSCP G6 and need not be adjusted.

The factory default COM Port setting is COM1. At start-up, if *NovaSource G6* is not found on this port (or at the currently specified port), an error message displays. In this case, you must use the Settings Menu, Comm Port... to choose another port. The port change will be saved as your default.

Help



The Help Menu launches the *NovaSource G6* User Manual and provides information about the NSCP G6.

Selecting and Changing Controls

To select a control to change in the Main Control Panel Window or in the Sweep Window, click on the control and press the TAB key repeatedly until the appropriate value is highlighted, or hold down the ALT key while pressing the hot-key (indicated in the Control Label with an underscore).

When a numeric value is selected, it is highlighted to let you know the value can be erased by typing a new value. If a typed value is out of range for the particular control, a warning message is displayed in the Status Display at the bottom of the Control Panel

Window. If an invalid character is typed, an error message is displayed. In either case, the previous value is retained. Selection of control buttons and check boxes are indicated by a dotted line around the Control Label.

Typing a new value in a numeric control field changes the setting on the *NovaSource G6*. This change does not take place until the ENTER key is pressed or another object in the window is selected by clicking on it with the mouse or by pressing the TAB key. Changing the value of a control may affect the value of other controls.

Frequency Controls

The current setting of the *NovaSource G6* RF output frequency is shown in the Frequency control field. The units are shown to the right of the value and can be changed by double-clicking on the units to display the frequency in Hz, kHz, or MHz.

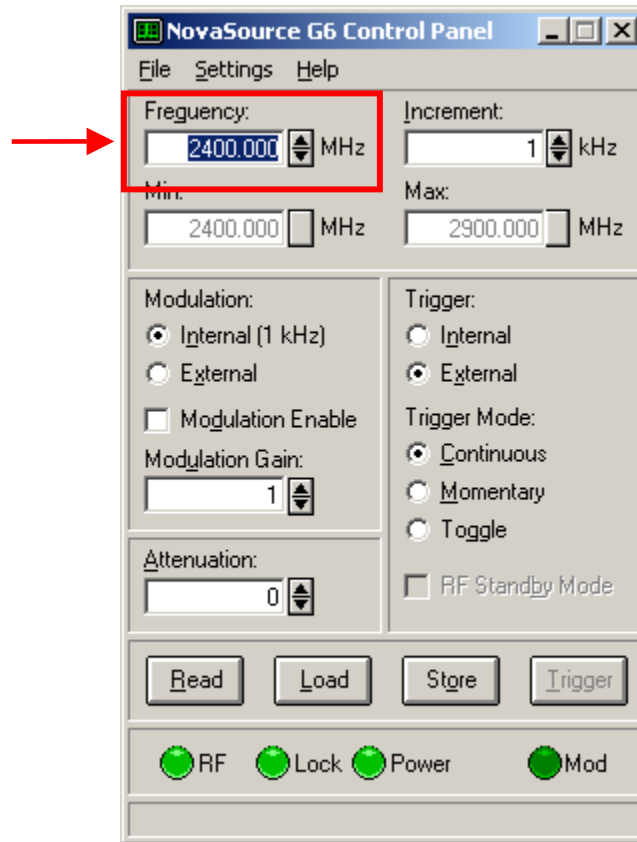


Figure 7: NovaSource G6 Control Panel Frequency Adjustment

A new frequency can be typed directly into the Frequency control when the control is selected. Entering or adjusting to a frequency outside of the Min and Max frequencies causes the NSCP G6 to issue a warning and retain the previous output frequency value.

Any value entered that is not a multiple of the Increment is rounded to the nearest multiple of the Increment.

While the cursor is in the Frequency control text box, the UP and DOWN arrows on the keyboard affect the control. UP increases the frequency; DOWN decreases the frequency. The amount of increase or decrease is determined by the value in the Increment control. For these two controls, holding down the SHIFT, CTRL, or ALT keys while pressing the UP and DOWN arrow keys increases or decreases the value of the control by 10, 100, or 1000 times the increment value, respectively, in order to allow faster tuning.

In addition, the Frequency and Increment controls can be changed by using the up or down arrows to the right of the control's value. Clicking on the UP arrow increases the control's value; clicking on the DOWN arrow decreases the control's value. The unit amount of increase or decrease can be changed by holding down the SHIFT, CTRL, or ALT keys while pressing the up and down arrows to increase or decrease the control's value by 10, 100, or 1000 times its unit value, respectively.

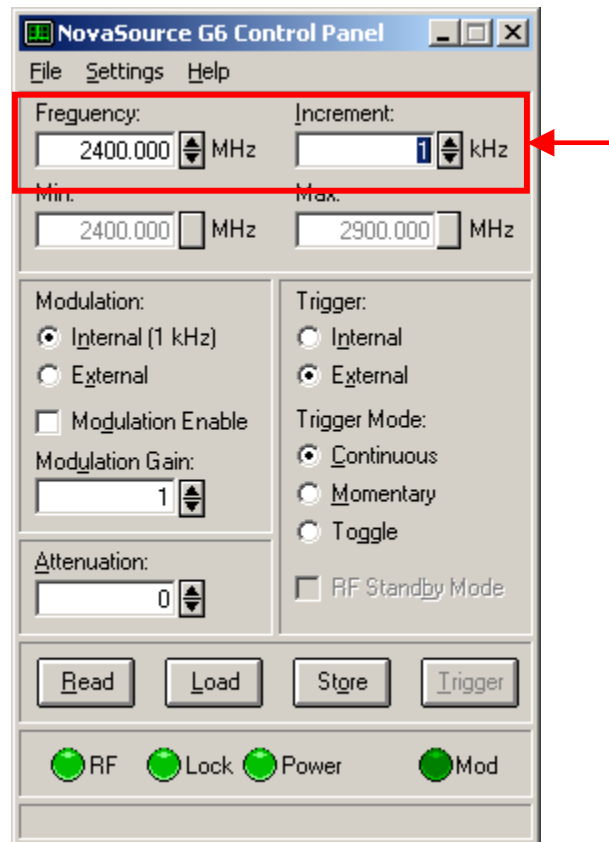


Figure 8: NovaSource G6 Control Panel Step Size Adjustment

Changing the Increment value has no effect on the output Frequency value until the Frequency is incremented or decremented. The Increment value must be at least 1 kHz and cannot be greater than the tuning range of the attached NovaSource G6. Attempting to set the Increment to an illegal value will cause the NSCP G6 to issue a warning message and default to the previous increment value.

The remaining two frequency controls in the Main Control Panel Window are display only. The values displayed are for informational purposes only and cannot be changed. The Min field shows the minimum output frequency that the attached *NovaSource G6* can support. The Max field shows the maximum output frequency that the attached *NovaSource G6* can support.

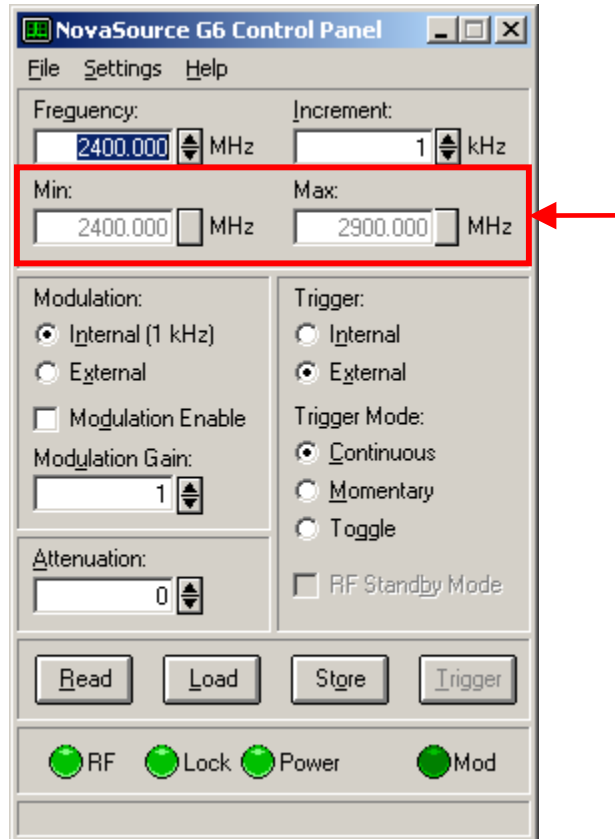


Figure 9: NovaSource G6 Control Panel Min/Max Fields

Attenuation

The Attenuation Control is used to display and modify the RF output attenuation level. Entering a new attenuation level or adjusting the value with the arrow keys or UP/DOWN buttons causes the RF output to be attenuated by the new attenuation level.

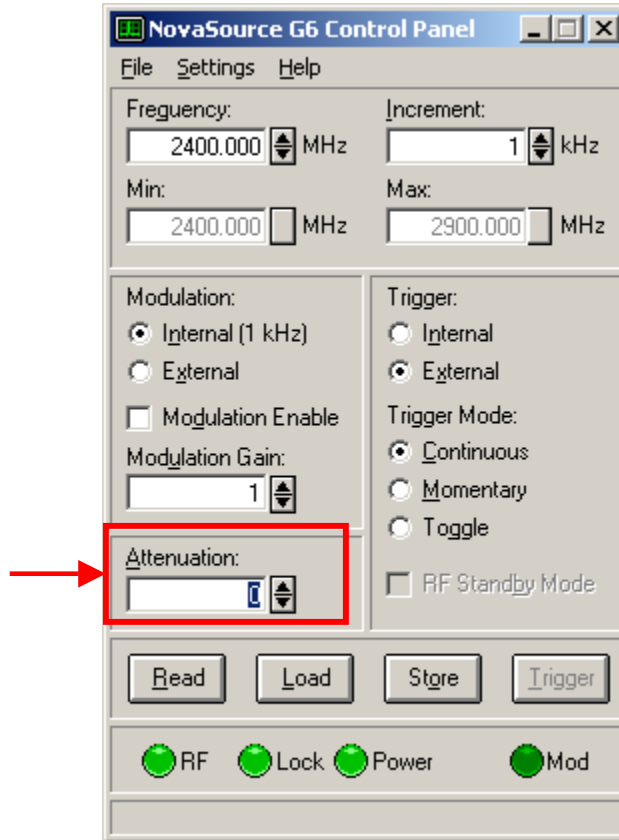


Figure 10: NovaSource G6 Control Panel Attenuation Control

The Attenuation Control range is 0 to 31. Entering or adjusting to a setting outside of that range causes the NSCP G6 to issue a warning and retain the previous RF output attenuation level.

The unit of increment and decrement used by the UP and DOWN arrow buttons and the Control Adjustment is always 1.

Trigger Control

There are five “option buttons” on the Control Panel window used to define the operation of the trigger for the *NovaSource G6*. The Trigger Source is selected using the Internal and External option buttons. In addition, the trigger can act in one of three ways: Continuous, Momentary, or Toggle.

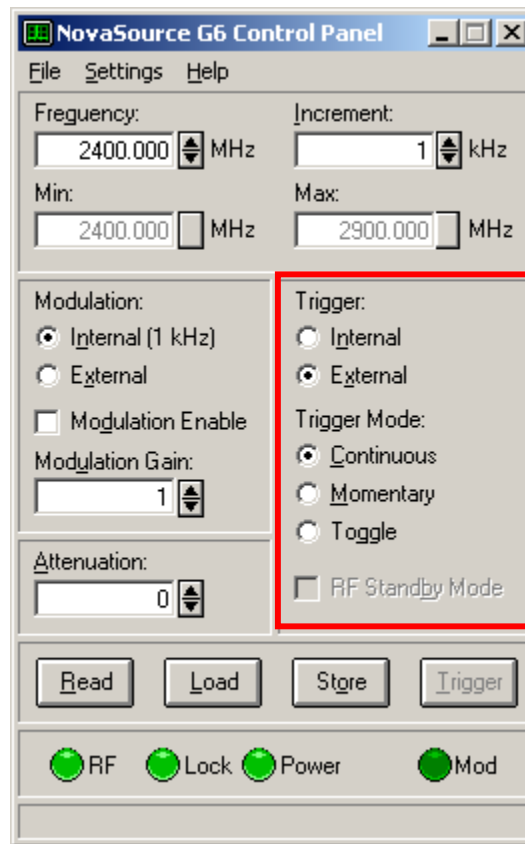


Figure 11: NovaSource G6 Control Panel Trigger Control

Selecting the Trigger Source

The RF output can be triggered by either the internal control or an external signal. The internal control is provided on the NSCP G6 control panel by the TRIGGER function button. This button is only enabled when the Internal trigger source is selected.

To use an external trigger source, click on the External trigger source button. When this option button is selected, the *NovaSource G6* will respond to an external signal applied at the TRIG/MOD input connector. When acting as the trigger source, this input is a 5 volt CMOS input and is active low. That is, the RF output will be ON when the input is LOW.

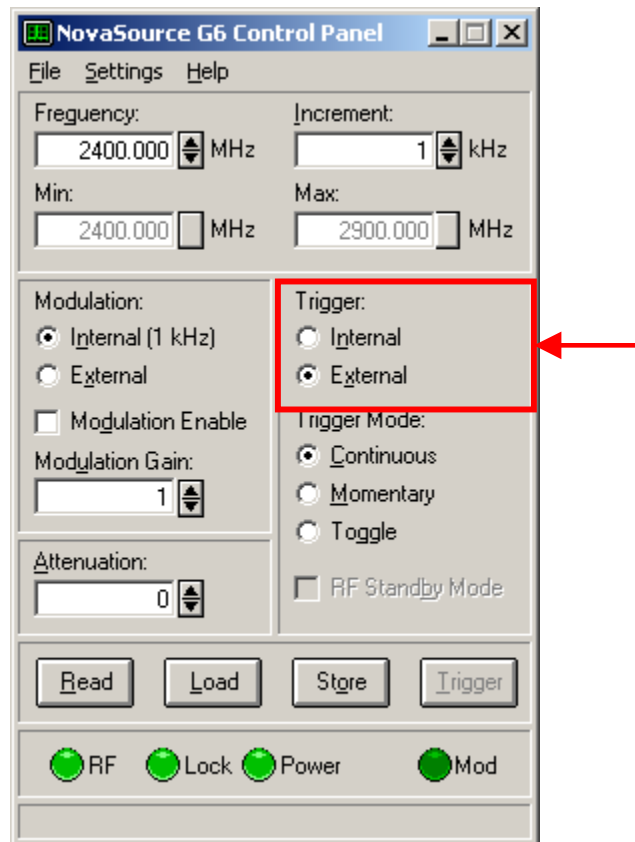


Figure 12: NovaSource G6 Control Panel Trigger Source Options

Continuous, Momentary, Toggle

Selecting Continuous trigger mode puts *NovaSource G6* into continuous RF output mode. In this mode the trigger, either internal or external, has no effect on the RF output. (RF output is always on.)

Momentary puts *NovaSource G6* into momentary RF output mode. The RF output is turned on with the falling edge of the trigger and turned off with the rising edge of the trigger.

When Toggle mode is selected, the falling edge of the trigger turns the RF output on (if it was off) and off (if it was on). Momentary and Toggle enable the Trigger button (refer to the section below). Continuous disables the Trigger button.

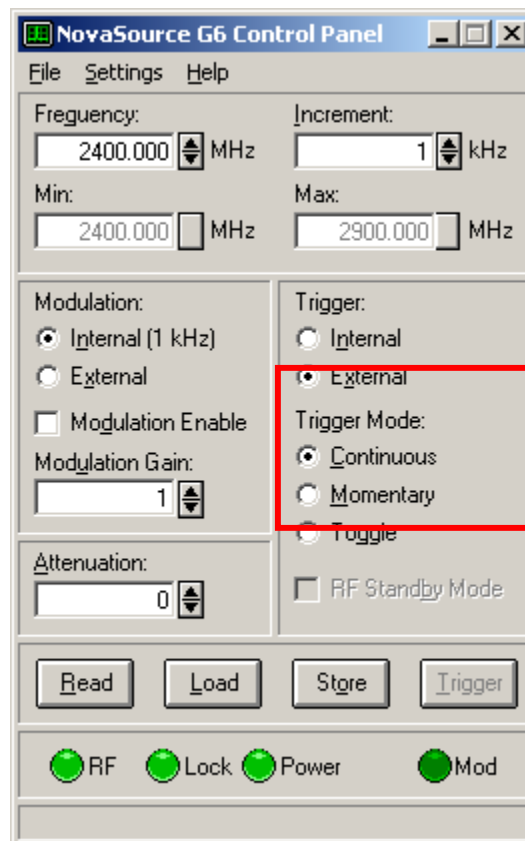


Figure 13: NovaSource G6 Control Panel Trigger Mode Options

RF Standby Mode

The RF Standby Mode “checkbox” is used to control *NovaSource G6* RF standby mode. RF standby mode affects the RF output signal in two ways: RF On lockup time, and RF Off output level.

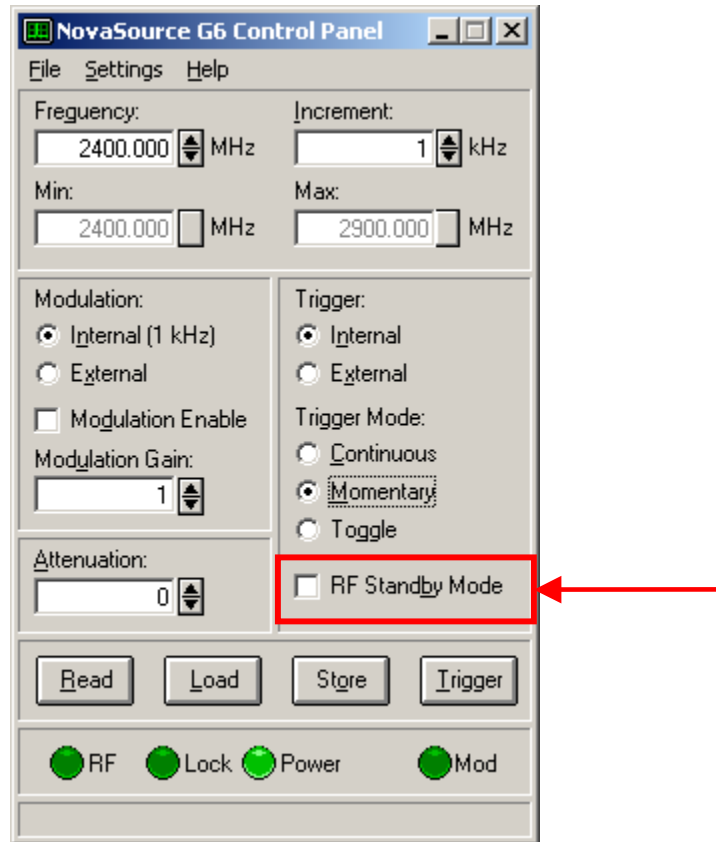


Figure 14: NovaSource G6 Control Panel RF Standby Mode Active

When RF Standby Mode is checked, the *NovaSource G6* circuitry is controlled in a way that allows the output to be turned on and off rapidly, but the output level when off is not as low as when Standby is not checked. Applications that require high-speed testing should select RF Standby Mode as long as the low level output present when the *NovaSource* is off does not cause problems in the tests. If it is important that the *NovaSource* signal be completely off when not in use, RF Standby Mode should not be used.

Function Buttons

There are four “buttons” on the Control Panel window. These buttons are independent and perform different functions.

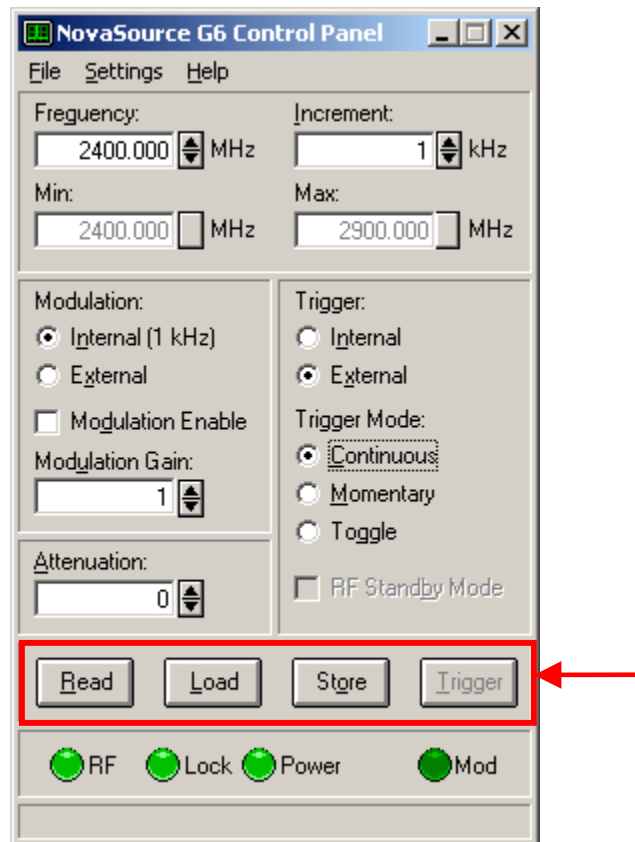


Figure 15: NovaSource G6 Control Panel Function Buttons

Read

The Read Button causes NSCP G6 to interrogate the attached NovaSource G6. The Frequency, Attenuation, Trigger, Modulation, and RF Standby Mode displays are updated as a result of the returned values.

Load and Store

NovaSource G6 retains the configured output frequency and all other settings when power is removed. When *NovaSource G6* is powered on, the “stored” configuration becomes the “current” configuration. Any value changed via NSCP *G6* affects only the current configuration and not the stored (power-on) configuration until you click on the Store Button. Until a configuration is “stored,” it will be lost if power to the unit is removed. The last stored configuration will remain the default.

The Load Button retrieves the stored configuration to the current configuration and updates NSCP *G6*. The Store Button copies the current configuration to the stored configuration (in other words, the current values become the power-on values).

Trigger

The Trigger Button controls the Internal trigger signal to *NovaSource G6*. Clicking on and holding the button makes the Internal trigger active; releasing it makes the trigger inactive. This button is not enabled when in Continuous Trigger Mode.

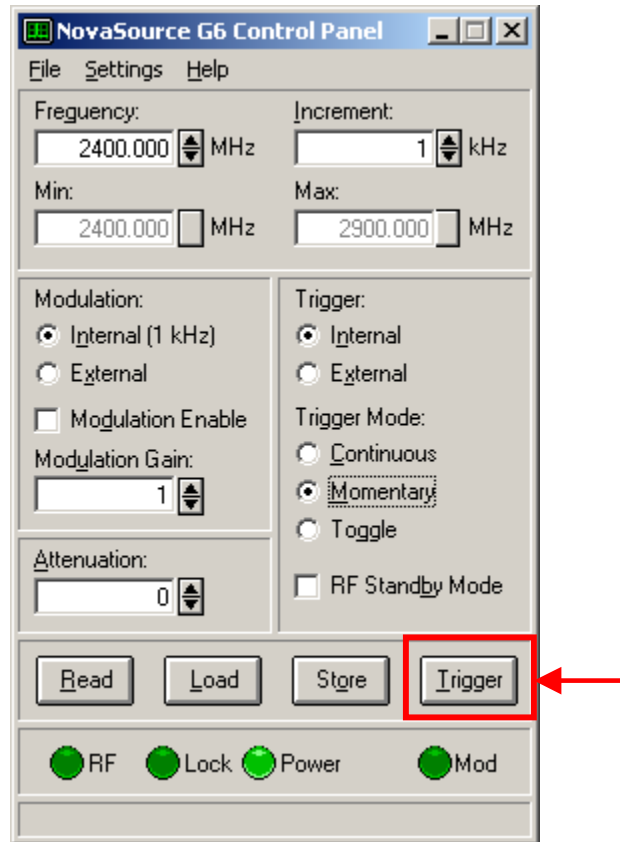


Figure 16: NovaSource G6 Control Panel Trigger Button

LED Display

The LED display corresponds to the front panel LEDs on the NovaSource G6. During normal operation, the NSCP G6 software will periodically poll the attached NovaSource G6 to update the LED states.

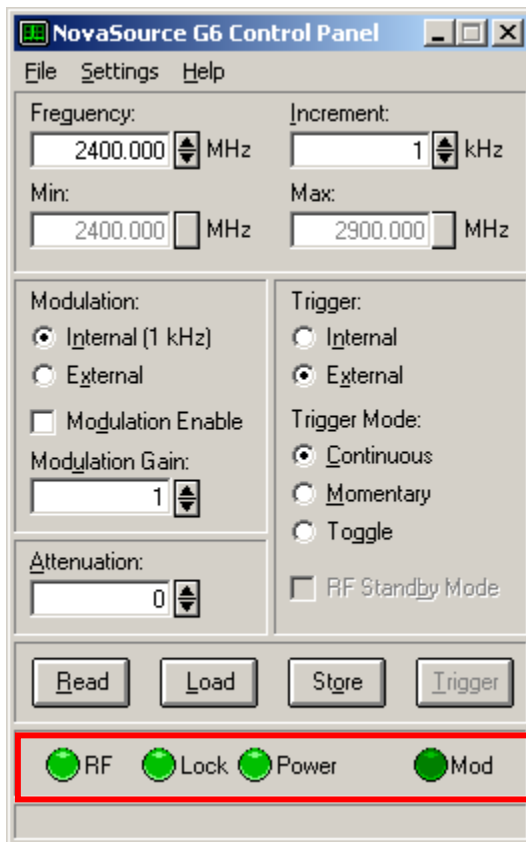


Figure 17: NovaSource G6 Control Panel LED Display

Status Display

The Status Display is a single line text window located below the LED Display. It is used to display warning messages and show the status of various events.

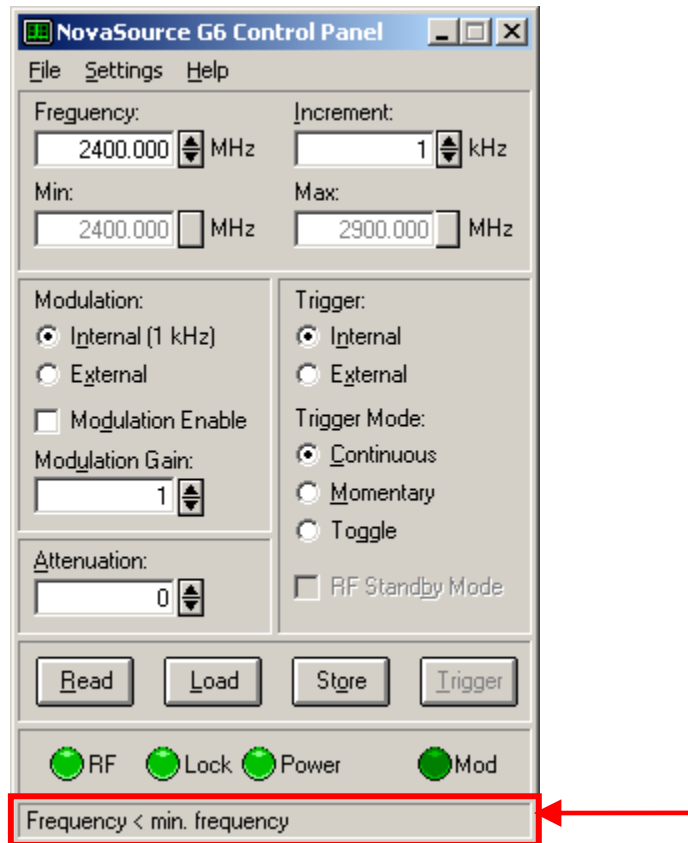


Figure 18: NovaSource G6 Control Panel Status Display

Modulation Controls

The NovaSource G6 supports modulation of the RF output from an internal 1 kHz source or from an externally supplied modulation signal. Note that the externally supplied signal must be applied to the TRIG/MOD input connector, so that the external trigger function is not available when using external modulation.

Modulation Source Selection

Two option buttons are used to select the modulation source. To select the internal 1 kHz source, click on “Internal.” To select the external input, click on “External.” In addition to these choices being mutually exclusive, the external modulation input is not available if the TRIG/MOD input is being used as the input for an external trigger. An error message is displayed in the Status Window if this is attempted. You must select the internal trigger in order to use the TRIG/MOD input for modulation.

Modulation On/Off Control

Either modulation source is turned on and off using the Modulation Enable checkbox.

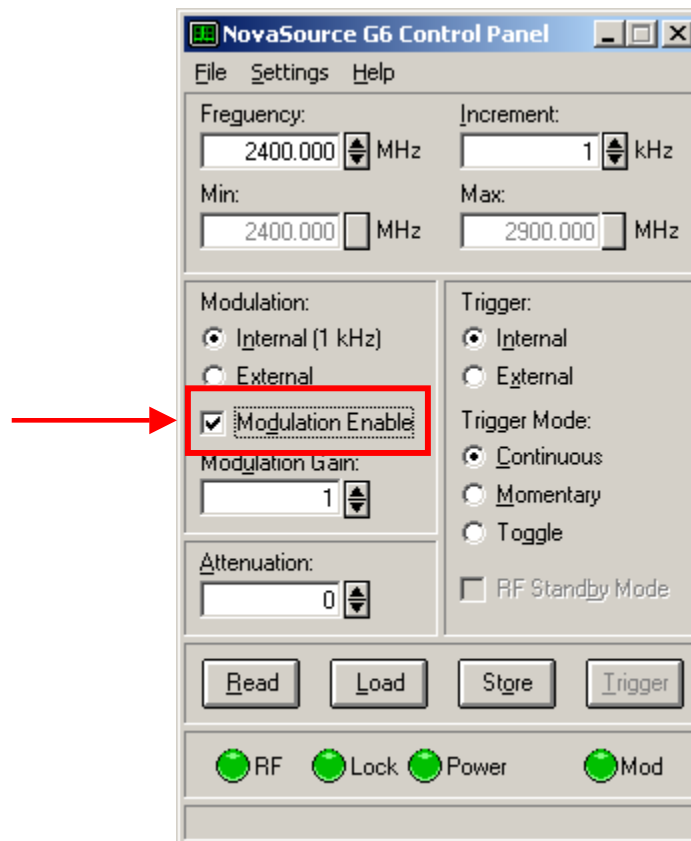


Figure 19: NovaSource G6 Control Modulation Enable Checkbox

When this box is checked and Internal modulation is selected, the RF output signal is FM modulated at a rate of 1 kHz. When this box is checked and external modulation is selected, the RF output is modulated by the signal applied to the TRIG/MOD input connector.

When the checkbox is unchecked, the RF output is not modulated and returns to the programmed center frequency regardless of the voltage at the TRIG/MOD input.

Deviation Control

The deviation that is imposed on the RF output by the modulation signal (internal or external) can be varied using the Modulation Gain setting.

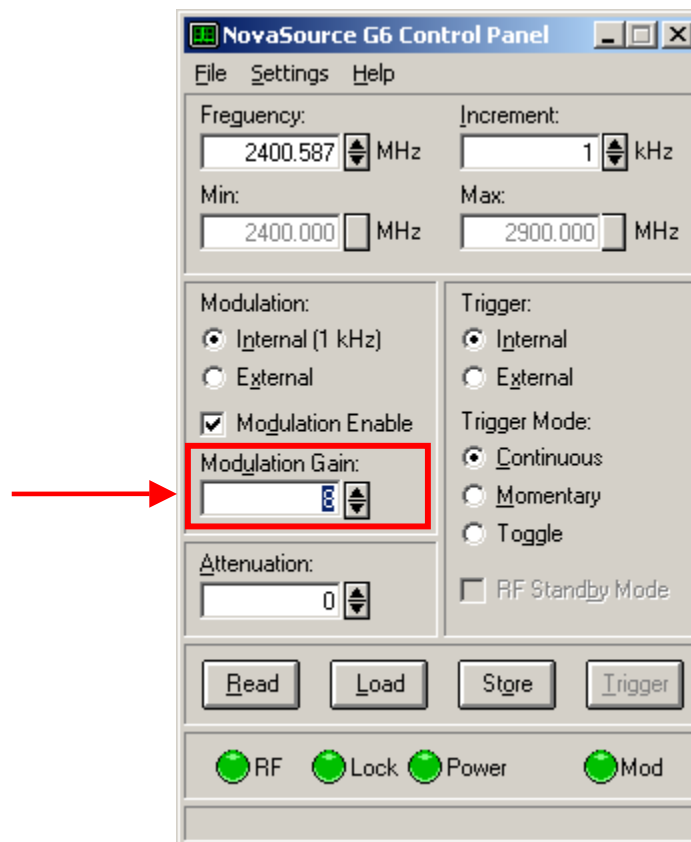
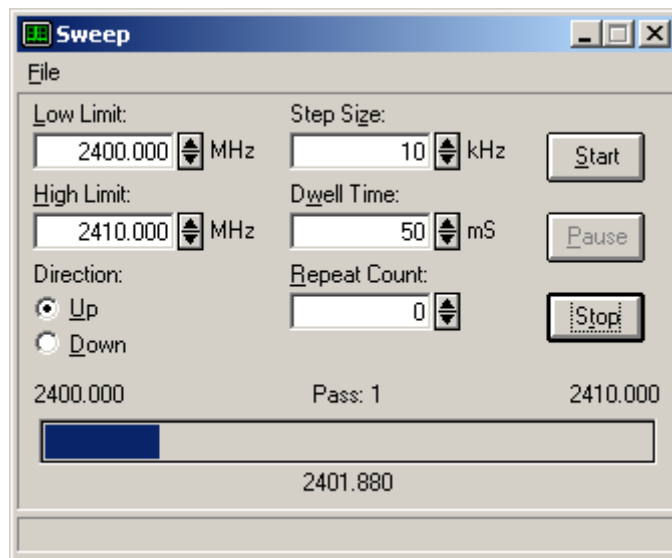


Figure 20: NovaSource G6 Control Panel Modulation Gain Control

The modulation signal is multiplied by this value before being applied to the PLL tuning data. Gain factors both greater than 1 and less than 1 are available, so that the modulation signal can be amplified and reduced (deviation increased and decreased) using this value.

Sweep Window

The Sweep Window is accessed from the Settings Menu and is used to generate sweeps between two end frequencies with user selectable step size, direction, and dwell time. Note that the Sweep Window uses standard *NovaSource G6* commands to create sweeps from within the NSCP G6 software. The sweep functionality does not reside within the *NovaSource G6*. For instructions on creating sweeps from user applications, refer to the Application Note “Programming the *NovaSource G6*” available by contracting Nova Engineering or from our website <http://www.nova-eng.com>.



Field Name	Description
Low Limit	The lowest frequency to be included in the frequency sweep
Step Size	Size in kHz of each step in the frequency sweep
High Limit	The highest frequency to be included in the frequency sweep

Dwell Time	Time in milliseconds between frequency steps
Direction: Up/Down	Up—Sweep from the Low Limit frequency to the High Limit frequency Down—Sweep from the High Limit frequency to the Low Limit frequency
Repeat Count	Number of times to run the frequency sweep (0 = continuous)
Start Button	Click on START to start the sweep.
Pause Button	Click on PAUSE to temporarily interrupt the sweep; the button grays out. Click on START to continue the sweep.
Stop Button	Click on STOP to halt the sweep.
Progress Box	The blue progress line indicates the status of the sweep. From left to right, the line increments from 0 to 100 percent complete. The number below the Progress Box displays the frequency as the sweep advances.

To define a sweep, complete the following steps:

- Select the Low Limit field and type the lowest frequency to be included in the sweep. This frequency must be within the tuning range of the attached *NovaSource G6* and must be lower than the High Limit.
- Select the High Limit field and type the highest frequency to be included in the sweep. This frequency must be within the tuning range of the attached *NovaSource G6* and must be higher than the Low Limit.
- Select the Step Size field and type the frequency step to be taken between adjacent frequencies in the sweep. The step size is in kHz and can be as small as 1 kHz or as large as the entire sweep range (Low Limit to High Limit).
- Select the Dwell Time field and enter the time, in milliseconds, that the *NovaSource G6* should stay at each frequency. This value can range from 50 ms to 60000 ms (60 seconds).
- Select the Repeat Count field and enter the number of times that the sweep from Low Limit to High Limit should be performed. For continuous sweeping, enter zero.
- Click on either the Up button or the Down button to define whether the sweep is performed with increasing or decreasing frequency.

- The sweep is now defined. Press Start to begin the sweep.

After the sweep is started, it can be stopped by clicking on the Stop button. When stopped in this manner, clicking Start will begin the sweep again from the original starting frequency. The sweep will then repeat as set in the Repeat Count field.

To stop the sweep only temporarily, click on the Pause button. This button will gray out and the focus will shift to the Stop button. Clicking on the Start button causes the sweep to resume from the frequency at which it was stopped. Clicking Stop while paused causes the sweep to end and will only restart from the original starting frequency.

As the sweep progresses, the progress bar at the bottom of the Sweep Window indicates how close to completion the sweep is. The value under the bar shows the exact frequency to which the *NovaSource G6* is tuned.

For added flexibility, the Step Size, Dwell Time, and Direction can be changed while a sweep is active. This allows the sweep to be slowed, reversed, or changed to a finer resolution in order to examine parts of the spectrum as desired.

How NovaSource G6 Works

NovaSource G6 provides a frequency-programmable low phase-noise RF signal. This is achieved by employing a Phase Locked Loop (PLL). In the PLL a voltage controlled oscillator (VCO) generates an RF signal that is phase locked to a low phase noise temperature compensated crystal oscillator (TCXO) reference signal. By phase-locking the VCO to the crystal reference, the VCO-generated RF signal takes on the good qualities of the crystal oscillator, such as lower phase noise and excellent frequency stability.

A PLL operates as follows. The TCXO signal is divided down to the comparison frequency. The VCO's output is also divided down to the comparison frequency using a programmable divider, which we will call the main divider. With both signals divided down to a common comparison frequency, they are both fed into a Phase Detector (PD). The PD output is fed back to the VCO and increases or decreases the VCO frequency to keep the divided-down VCO and TCXO signal in phase lock.

Three "trigger" modes are available for controlling the RF output state via the external trigger input: continuous, momentary, and toggle. In continuous mode, RF output is on "continuously" and is unaffected by a trigger event. In momentary mode, the RF output is on only while the trigger is active. In toggle mode, the RF output is alternately turned on or off by a trigger event.

NovaSource G6 employs an RF standby mode to provide quicker on/off switching of the RF output signal. RF output on/off switching is controlled by the host or by an external trigger input when in momentary or toggle mode, and is accomplished by switching power to the voltage-controlled oscillator (VCO) on and off. When in RF standby mode, the VCO is left on and RF output on/off switching is accomplished by attenuating the output signal. This results in a quicker RF On response, as well as a less attenuated RF Off output signal level.

NovaSource G6 is capable of FM modulation including FSK, MSK, and GMSK via the external TRIG/MOD port. When external modulation is enabled, the signal applied here is

digitized and applied to the PLL tuning data, causing the RF output frequency to track the modulation signal. Internal modulation with a 1 kHz internal sine wave is also available.

Power

NovaSource G6 requires 9 VDC @ 300 mA. A 9 VDC @ 500 mA wall plug power supply is supplied with North American models of *NovaSource G6*. Units sold outside of North America must be provided a suitable power source. A spare power supply connector is provided for use with an alternate power supply.

Note: When using a power supply other than the one provided by Nova Engineering, care should be taken to verify the proper output level prior to applying power to *NovaSource G6*.

About Nova Engineering

Founded in 1989, Nova Engineering creates state-of-the-art communications systems for military, government and industrial customers. With design and manufacturing facilities in Cincinnati and Largo, Florida the company employs nearly 150 people.

In October 2006, Nova was acquired by L-3 Communications and reports directly to Communications Systems-West (CS-West) in Salt Lake City.

Nova is a proven leader in advanced communication systems for customers such as the U.S. Army and Marines, as well as prime government contractors like Northrup Grumman, Lockheed Martin, Boeing, General Dynamics, and Textron.

Headquartered in New York City, L-3 Communications is a leading provider of Intelligence, Surveillance, and Reconnaissance (ISR) systems, secure communications systems, aircraft modernization, training, and government services. The company is a leading supplier of a broad array of high technology products, including guidance and navigation, sensors, scanners, data links, propulsion systems, simulators, avionics, electro optics, satellite communications, electrical power equipment, encryption, signal intelligence, antennas, and microwave components.

To learn more about L-3 Communications, please visit the company's web site at www.L-3Com.com. For more information about Nova Engineering, visit www.nova-eng.com or call 513-642-3000.

Getting Help

Technical Support

- World Wide Web: <http://www.nova-eng.com/>
- General E-mail: info@nova-eng.com
- Sales Email: sales@nova-eng.com
- FTP Server: <ftp://nova-eng.com/>

- Voice: +1-513-642-3000
- Fax: +1-513-642-3300

Before calling L-3 Communications Nova Engineering, please have the following information ready:

- A description of the problem
- A description of any action(s) already taken to resolve the problem
- The hardware serial number, firmware and software version number
- Application of product
- The device history (i.e., have you returned the device before, is this a recurring problem, etc.)

Copyright

© 2007 L-3 Communications Nova Engineering, Cincinnati, Ohio, USA. All rights reserved. This manual and the software described in it are copyrighted with all rights reserved. No part of this publication may be reproduced, transmitted, transcribed, stored in a retrieval system, or translated into any language in any form by any means without the written permission of L-3 Communications Nova Engineering.

L-3 Communications Nova Engineering reserves the right to make changes in specifications and other information contained in this document without prior notice. The hardware, firmware, or software described in this manual is subject to change without notice.

In no event shall L-3 Communications Nova Engineering be liable for any incidental, indirect, special, or consequential damages whatsoever (including, but not limited to, lost profits) arising out of, or related to, this manual or the information contained in it, even if L-3 Communications Nova Engineering knew of, should have known, or has been advised of, the possibility of such damages.

Trademarks

NovaSource G6, *NovaSource G6 Control Panel*, and the L-3 Communications Nova Engineering logo are trademarks of L-3 Communications Nova Engineering. All other trademarks are the property of their respective owners.

Nova Engineering Product Warranty

Nova Engineering, Inc. warrants that its products shall, upon delivery, be free from defects in workmanship or material. Should any failure of the to conform to this warranty appear within twelve (12) months from the date of delivery, Nova Engineering shall, provided that it is given immediate written notice of such nonconformity and provided that Nova Engineering's inspection confirms the existence of same, correct such nonconformity by, at its own election, repair or replacement, Free On Board (FOB) Nova Engineering's plant, or refund of the purchase price of the nonconforming product or part.

This warranty shall be null and void if there has been any attempt to tamper with, modify, or repair the product. This warranty does not cover and Nova Engineering will not be liable for any damage or failure caused by misuse, abuse, acts of nature, accidents, or other causes beyond Nova Engineering's control, or claim by other than the original purchaser.

Repair, replacement, or refund in the manner provided above shall constitute fulfillment of all Nova Engineering's obligations under this warranty. This warranty shall not apply to any product or parts that have been subject to accident, misuse, unauthorized alterations, or where Nova Engineering's installation and service requirements have not been met.

Return of product to Nova Engineering's facility for inspection and repair is the Buyer's responsibility. Repair of product under warranty may be refused if payment in full has not been received.

THIS WARRANTY IS EXCLUSIVE AND IS IN LIEU OF ANY WARRANTY OF MERCHANTABILITY, FITNESS FOR PURPOSE, OR OTHER WARRANTY OF QUALITY, WHETHER EXPRESS OR IMPLIED. IN NO EVENT SHALL NOVA ENGINEERING BE LIABLE FOR SPECIAL, INDIRECT, INCIDENTAL, OR CONSEQUENTIAL DAMAGES.

To obtain service under this warranty:

First, obtain a Return Material Authorization (RMA) Number by calling, faxing, or e-mailing Nova Engineering at the following:

Voice Phone: +1-513-642-3000

Fax +1-513-642-3300

E-mail support@nova-eng.com

You will be asked to provide the model number, serial number, and purchase date of your product.

Return the product (appropriately packed and freight prepaid) with a brief description of the problem to the following address:

Nova Engineering, Inc.

Attn: Repair, RMA# _____

5 Circle Freeway Drive

Cincinnati, OH 45246-1201

USA

NovaSource G6 Specifications

Table 2: General Specifications, All Models

Parameter	Performance			Units
Reference	Min.	Typical	Max.	
Frequency Reference Port		SMA jack; selectable as internal reference output or external reference input		
Internal Reference Frequency		10.00000		MHz
Initial internal reference initial frequency accuracy (@ +23 +/- 2 deg. C)	-1.0		+1.0	ppm
Initial internal reference temperature stability, over operating temperature range	-1.0		+1.0	ppm
Internal reference frequency aging		Less than +/- 1.0		ppm/yr.
Internal reference output level	+0	+2 1.5		dBm, 50 Ω load Vpk-pk, 1 M Ω load
External reference frequency (dependent on PLL configuration)		Up to 40		MHz
External reference input level		0.6 to 5.0		Vpp
External reference input impedance		600 Ω in parallel with 25 pF		n/a
RF Output				
RF output power (highest step)	+5.0	+8.0		dBm
RF output power adjustment		Software Controlled		
Number of steps		31		
Total Range	25	28		dB
Non-harmonic spurious		<-60		dBc
Harmonics		<-35		dBc
RF output modes		Continuous, Momentary, and Toggle (on/off)		
RF OFF output level (RF Standby not enabled)		VCO powered off – no output		

Parameter	Performance			Units
Reference	Min.	Typical	Max.	
Lock time (RF Standby <i>not enabled</i>): Ext. trigger falling edge RF within 30 degrees of final phase		<35 (most models <15)		msec
RF OFF output level (RF Standby enabled)		-70 (45 - 65 MHz model) to -40 (2000 - 2500 MHz model)	-40	dBm
Lock time (RF Standby <i>enabled</i>): Ext. trigger falling edge to RF within 30 degrees of final phase		<6.0 (most models <2.0)		msec
External Trigger				
Input Range	0		+5	VDC
Polarity		Active Low		
Modulation				
FM		Internal 1 kHz rate, or external		
Rate			50	kHz
Peak Deviation		±1.25 MHz using 10 MHz reference		MHz
External Input Range	-2.5		+2.5	VDC
Power Supply				
DC supply input voltage	+9		+15	VDC
DC supply current	175		300	mA
Environmental / Mechanical				
Temperature range, operating	0		+50	deg. C
Temperature range, storage	-20		+70	deg. C
Control Interface		RS-232 serial		
Control Software		PC – Windows 95/98/2000/XP compatible		
Status Indicator LEDs		RF Output ON Phase Lock DC Power applied		
Size (W x L x H)		2.75" x 4.0" x 0.75 70 mm x 102 mm x 19 mm		Inches mm
Weight (not including cables or power supply)		6 170		oz. g.
RF Output connector		SMA female		
Reference connector		SMA female		
Trigger / Modulation connector		SMA female		

Parameter	Performance			Units
Reference	Min.	Typical	Max.	
Power connector		1.3 mm center-positive DC power jack		
Data connector		DB-9 female		
Enclosure		Extruded aluminum, plastic end-caps		
Finish		Epoxy-based paint, white		

Table 3: Additional Specifications by Model Number

(Frequency range and step size guaranteed to within reference frequency accuracy.)

NovaSource G6 Model Number	Frequency Range (MHz) (min-max)	Step Size (kHz)	SSB Phase Noise (dBc/Hz) (all typical)
NS3-2400102	2400-2900	1 kHz	-69.0 @ 1 kHz removed -76.0 @ 10 kHz -105 @ 100 kHz
NS3-3400102	3400-3700	1 kHz	-70.0 @ 1 kHz removed -78.0 @ 10 kHz -109 @ 100 kHz
NS3-3700102	3700-4200	1 kHz	-67.0 @ 1 kHz removed -75.0 @ 10 kHz -103 @ 100 kHz
NS3-5150102	5150-5350	1 kHz	-67.0 @ 1 kHz removed -75.0 @ 10 kHz -103 @ 100 kHz
NS3-5470102	5470-5875	1 kHz	-61.0 @ 1 kHz removed -71.0 @ 10 kHz -96 @ 100 kHz

Other NovaSource models are available. For information, visit Nova's website at <http://www.nova-eng.com> or contact our Sales department.

Index

Attenuation, 20
Continuous, Momentary, Toggle, 23
Control Panel Window, 10, 11
File Menu, 12
Frequency Controls, 17
Function Buttons, 26
Help, 16
How NovaSource G6 Works, 36
Installing NovaSource G6, 6
LED Display, 28
Load and Store, 27
Menus, 12
Nova Engineering, 39
NovaSource G6 Specifications, 44

NSCP G6, 10
Power, 38
Product Warranty, 42
Read Button, 26
RF Standby Mode, 24
Selecting and Changing Controls, 16
Set Up, 7
Settings, 14
Status Display, 30
Technical Support, 39
Trigger Button, 27
Trigger Control, 22
Warranty, 42



L-3 Communications Nova Engineering

5 Circle Freeway Drive
Cincinnati, OH 45246-1201 USA

1-800-341-NOVA (6682)

+1-513-642-3000
FAX +1-513-642-3300

www.nova-eng.com
info@nova-eng.com

Part #: NL-NS002-020315 *NovaSource G6 RF Signal Source* User Manual

Revision: 24010712

© 2007 L-3 Communications Nova Engineering. All Rights Reserved.